

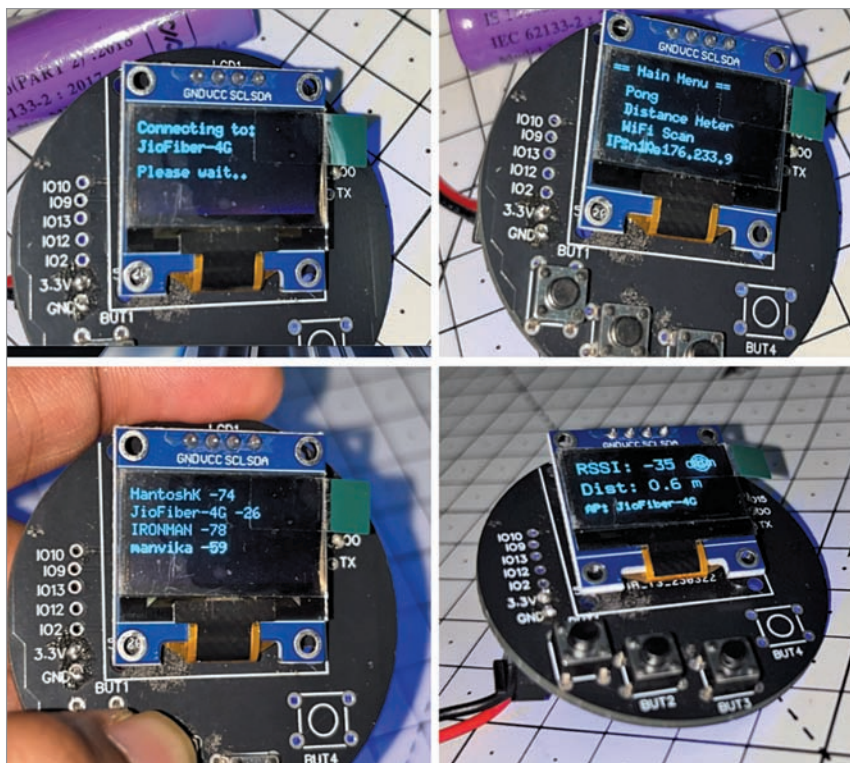


Fig. 5: Testing of the device's various functions

mounted on the bottom side. This arrangement allows the device to be rotated with the thumb while the controls are operated with the same hand.

The board supports both the SSD1306 OLED and GC9A01 colour TFT displays, allowing either display to be fitted without redesigning the PCB. Fig. 3 shows the top and bottom PCB layouts and their corresponding 3D views of the Wi-Fi tool and indoor location tracking device.

Software and code

The software is developed for the ESP32-S3 using the Arduino framework and Arduino IDE. The program uses the WiFi.h library for Wi-Fi connectivity, ESPmDNS.h for network name resolution, and ArduinoOTA.h for over-the-air firmware updates. It supports two display options: an SSD1306 128 × 64 OLED using the I²C interface through the U8g2lib.h library, and a GC9A01 colour TFT using the SPI interface through the TFT_eSPI.h library. Conditional

compilation using USE_OLED or USE_TFT allows either display to be selected at compile time. The Wi-Fi SSID and password are defined in the configuration section and used by the ESP32-S3 to establish a network connection.

The code also defines the GPIO pins for the user interface according to the circuit diagram. Buttons SW2, SW3, SW4, and SW5 are connected to IO13, IO14, IO35, and IO36, respectively, and are used for menu navigation and selection. SW1 is connected to IO0 for boot/programming control, while the 50kΩ potentiometer is connected to IO5 to provide an analogue input for

user control. After startup, ESP32-S3 initialises the selected display, connects to the Wi-Fi network, and performs the required sensing and signal-processing functions. Fig. 4 shows the code snippet for configuring Wi-Fi.

The processed Wi-Fi signal and CSI information can then be displayed on the selected OLED or TFT screen, while the switches and potentiometer provide user input. The OTA library allows firmware to be updated wirelessly without a physical connection to the device.

Construction and testing

After all the components are assembled on the PCB, a battery can be connected to the battery-connector solder pads to power the device. The OLED initially displays 'Trying to connect' while the ESP32-S3 establishes a Wi-Fi connection. Once the connection is established, the main user interface appears, displaying a menu of available functions.

The up and down buttons are used to navigate through the menu and select the required function. The device provides options for viewing the CSI report, scanning nearby Wi-Fi networks, measuring the approximate distance from a selected access point, and performing Wi-Fi analysis. During testing, the SSD1306 OLED successfully displayed the Wi-Fi connection status, menu options, detected networks, RSSI values, and estimated distance. An RSSI of -35dBm corresponded to an indicated distance of approximately 0.6m in the test setup.

The four push-buttons were tested for menu navigation and selection, while the potentiometer provided additional user input. These tests confirmed the operation of the assembled PCB, OLED display, user controls, Wi-Fi connectivity, and sensing functions, as shown in Fig. 5. **EFY**

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